

We want to generate samples of a continuous r.v. X with cdf F_X .

Let U be a uniform r.v. over $[0,1]$.

Consider a r.v. $\hat{X}$ s.t.

$$\hat{X} = F_X^{-1}(U) \quad U = F_X(\hat{X})$$

Then, cdf of $\hat{X}$ is F_X .

$$F_{\hat{X}}(x) = P(\hat{X} \leq x) = P(F_X^{-1}(U) \leq x) = P(U \leq F_X(x)) = F_X(x)$$

$P(U \leq k) = k$

$\therefore$ Draw $u \sim U$ and obtain $F_X^{-1}(u)$.

This is a sample from $\hat{X}$ which has the same distribution as X .

Lemma: $\hat{X} = F_X^{-1}(U)$ has distribution $F_X(\cdot)$.

A consequence of this lemma is that $F_X(\hat{X}) = F_X(X)$ is a uniform distribution.
universality of uniform / probability integral transform

e.g. If you want to convert exponential to gaussian :

1. Convert exponential to uniform

$$U = F_X(X)$$

2. Convert uniform to gaussian

$$X = F_X^{-1}(U)$$

• Evaluating integrals via Monte Carlo approach

Suppose you want to compute $\theta = \int_0^1 g(x) dx$ using only samples from $U[0,1]$.

$$\theta = E[g(U)]$$

$$E[g(u)] = \int_0^1 g(u) \cdot \underbrace{f_U(u)}_1 \cdot du = \int_0^1 g(u) \cdot du$$

• Strong Law of Large Numbers:

$$\text{Let } X_i \text{ be iid and } S_n = \sum_{i=1}^n X_i$$

$$\text{Then, } \frac{S_n}{n} \longrightarrow E(X)$$

$\therefore$ Use iid samples of U and invoke SLLN.

$$\text{As } n \rightarrow \infty, \text{ we have } \sum_{i=1}^n \frac{g(U_i)}{n} \longrightarrow E[g(U)] = \theta$$

• Importance Sampling

Suppose you want to compute $E[h(X)]$.

You do not have access to samples from X but know its pdf $f(\cdot)$.

You have access to samples from another r.v. Y with pdf $g(\cdot)$.

$$\begin{aligned} E[h(X)] &= \int h(x) \cdot f(x) \cdot dx \\ &= \int \frac{h(y) \cdot f(y)}{g(y)} \cdot g(y) dy \\ &= E_Y \left[\frac{h(Y) \cdot f(Y)}{g(Y)} \right] \end{aligned}$$

Now use LLN and samples of Y to estimate $E[h(X)]$.

• Accept Reject Method

Suppose you want to generate samples of X with pmf $p(\cdot)$ using samples of Y with pmf $q(\cdot)$.

$$\text{Suppose } \frac{p(y)}{q(y)} \leq c \text{ for all } y \Rightarrow \frac{p(y)}{c \cdot q(y)} \leq 1$$

1. Generate a sample $y \sim q(\cdot)$
2. Generate a sample $u \sim U(0,1)$.
3. If $u \leq \frac{p(y)}{c \cdot q(y)}$, accept y as a sample from X .
If not, reject y and go back to step 1.

PROOF OF ACCEPT-REJECT METHOD:

$$P(\text{accept} | y) = P\left(u \leq \frac{p(y)}{c \cdot q(y)}\right) = \frac{p(y)}{c \cdot q(y)}$$

$$P(\text{sample } y \text{ and accept}) = q(y) \cdot \frac{p(y)}{c \cdot q(y)} = \frac{p(y)}{c}$$

$$P(\text{accept}) = \sum_y P(\text{sample } y \text{ and accept}) = \sum_y \frac{p(y)}{c} = \frac{1}{c}$$

$$P(y | \text{accept}) = \frac{P(\text{sample } y \text{ and accept})}{P(\text{accept})} = \frac{\frac{p(y)}{c}}{\frac{1}{c}} = p(y)$$

$$p(y | \text{accept}) = p(y)$$

$\therefore$ The accepted samples are distributed according to $p(\cdot)$.